PRESIDENTS OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MBEYA CITY COUNCIL

DR TULIA ACKSON SECONDARY SCHOOL

FORM ONE ASSESMENT EXAMINATION

041 **BASIC MATHEMATICS**

Time: 2: 30 hours **NOVEMBER, 2025**

INSTRUCTIONS

1. Answer all questions in these papers
2. All communication devices and any unauthorized materials are not allowed in the examination room.
3. Write your **FULL NAME** on every page of your answer sheet

FOR EXAMINER'S USE ONLY		
Question Number	**SCORE**	**EXAMINER'S INITIAL**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

EMAIL: IBNSHARIFFOMAR@GMAIL.COM PHONE: 0757-536-055

1. (a) Describe the concept of mathematics
   (b) Mathematics is all around us. Almost everything we do involve mathematics. Also, mathematics is related to other subjects in a very unique way. With vivid examples explain how each of the following subjects is connected to mathematics.
      (i) Physics (ii) Chemistry (iii) Biology
   (c) Explain five mathematical skills that are useful in your daily life activities.
   (d) Briefly explain two (2) areas where we apply mathematics

2. (a) Write down the rational number whose numerator is (-2×3) and whose denominator is $(23 - 16) \times (9 - 4)$.
   (b) A monthly salary of a certain worker is Tshs 250,000. One-fifth of his salary is spent in buying car fuels, half of remaining salary is spent in buying food and half of the rest is spent in miscellaneous expenses.
      (i) How much is his monthly savings?
      (ii) State whether the answer in part (i) is a rational number or not
   (c) Given that $m = 0.\dot{2}$ and $n = 0.0\dot{4}$, show that $m^2 = n\,(m + 1)$
   (d) (i) Convert $0.\dot{1}$ into fraction and hence find the value of $(0.\dot{1} \times 27)$
      (ii) Evaluate $0.\dot{2} + 0.\ddot{5}\dot{1}$ giving your answer as a fraction.

3. (a) Write each of the following repeating decimals by using a dot and a bar.
      (i) 0.3333... (ii) 0.639639639... (iii) 0.474747474…
   (b) (i) If $x = 2.\dot{6}$ and $y = 2.8\dot{3}$ find the value of $\frac{x}{y}$
      (ii) If $x = 0.\dot{3}$ and $y = 0.\dot{6}$ verify that $y^2 = x^2 + x$
   (c) The fraction, $a\ \frac{b}{c}$ is equivalent to $5.\dot{1}6\dot{5}$ Find the values of $a, b\ and\ c$.
   (d) A tank is full of water. When 60 liters are drawn off, it is $^2/_5$ full. What amount of water will the tank hold when it is full?

4. (a) Mwajuma has $\frac{1}{3}$ of a chocolate bar. She decides to share it equally among her 5 friends. How much chocolate will each friend get? Write your answer in fraction and decimal.
   (b) Show that $\frac{23}{9} = 2.55555$ … ….
   (c) Find an irrational number such that $x^2 = 7$
   (d) Given the numbers $-5, -3.5, 0, \frac{3}{4}, \sqrt{3}, \sqrt{5}, 9, -4.3\ddot{3}\ddot{5}, \pi, \frac{\sqrt{2}}{2}$. List the numbers which are:
      (i) Whole numbers (ii) Rational numbers (iii) Integers (iv) Irrational numbers

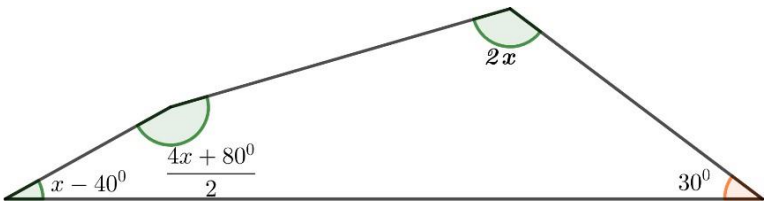
5. (a) If x is an integer, locate each of the following on a number line.
      (i) 1− is less than x and x is less than 4.
      (ii) 2− is less than or equal to x and x is less than or equal to 6.
   (b) In each of the following, insert a correct inequality symbols in the box to make a true statement.
      (i) $-5 \times (-3) \square\ 8 \times (-3)$ (b) $if\ a > b, then\ a \times (-9) \square\ b \times (-9)$
   (c) Anna has Tshs 500,000 in her bank account. Every week she withdraws Tshs 40,000 for various expenses. Without making any deposit, how many weeks can she withdraw this money if she wants to maintain a balance of at least Tshs 200,000?

(d) Write the inequalities represented by each of the following number lines.



6. (a) The XYZ cooperative society produces 94 jars of liquid soap every day. The cost of production of each jar consists of Tshs 4,015 for labour and Tshs 2,856 for materials. Estimate:
   (i) The total cost of producing a jar of liquid soap.
   (ii) The cost of materials for producing 94 jars.
   (iii) The cost of labour for producing 94 jars.
   (b) Write 47628.02415 correct to:
   (i) 3 significant figures (ii) 2 decimal places (iii) 3 decimal places (iv) 2 significant figures
   (c) A tablet contains 0.1763688 milligrams of a certain chemical. A dose requires a child to take $\frac{1}{2}$ a tablet, three times a day for 7 days. Find the total amount of chemical in the dose correct to hundredths of a milligram.
   (d) A certain boys' school with 2,485 students has decided to make uniforms for all students. If a shirt and a pair of trousers use 2 m and 1.5 m, respectively, how much materials of cloth in total will the school need? Give your answer to the nearest hundreds.

7. (a) A man gave $\frac{7}{25}$ of his money to his son, $\frac{2}{9}$ of the remaining amount to his daughter, and the rest amount to his wife. If his wife got Tshs 11,990,300;
   (i) How much money to the nearest ten thousand did the man have in total?
   (ii) Estimate the amount of money to the nearest hundred thousand collected by his daughter.
   (b) (i) A special cereal mixture contains rice, wheat, and corn in the ratio 2:3:5, respectively. If a bag of the mixture contains 3 kilograms of rice, how much corn does it contain?
   (ii) Find $(5x - 2y):(3x - y)$ in its lowest form if $x:y = 4:9$
   (c) (i) The denominator of a fraction exceeds the numerator by 5. If 4 is subtracted from the numerator and the denominator remains unchanged, the resulting fraction is $\frac{2}{5}$ Find the original fraction.
   (ii) A line segment AB, which is 24 cm long, is divided at a point P (between A and B) in the ratio 7:5. Find the lengths of $\overline{AP}$ and $\overline{PB}$
   (d) In the following figure, calculate the value of x in degrees, and find the ratio of two greatest angles.



8. (a) Simplify $8m^2n - 6a^2c + 4m^2n + 3a^2c - 8$ and then, state:
   (a) The coefficient of m^2n (b) The number of terms. (c) The constant term.
   (b) (i) Simplify of the following expressions: $(-10n)\left(-\frac{1}{10}\right) + \frac{1}{3}(9n) - \frac{1}{2}(-2n)$
   (ii) Show that $|x - y| = |y - x|$ for all real numbers x and y

(c) Determine the solution of each of the following inequalities.

(i) $\sqrt{2(x+1)} \geq 0$ (ii) $3(2x+1) \leq 2(x-1)$

(d) A certain school placed two orders with the stationery dealers. The first order was 50 boxes of marker pens and 20 boxes of correction fluid for the total amount of 40,000 Tanzanian shillings. The second order was 10 boxes of marker pens and 20 boxes of correction fluid for the total amount of 24,000 Tanzanian shillings.

(i) Formulate a system of simultaneous equations that describes the cost of a box of marker pen and a box of correction fluid.

(ii) Calculate the cost of a box of marker pens

9. (a) A line passes through points P(k, 4) and Q(3, k + 2). What value of **k** will make the line to be

(i) horizontal (ii) vertical?

(b) What should be the value of t for the straight line represented by the equation $3x - ty = 16$ to pass through the point (5, –4)?

(c) A line has an equation $ax + (a-1)y + 2a = 0$

(i) Rewrite the equation in the form of $y = mx + c$

(ii) If the line has y-intercept at (0, 2), find the value of a

d) (i) The points $(4, -2)$ and $(-2, -5)$ lie on the straight line $\frac{x}{a} + \frac{y}{b} = 1$ Find the values of a *and* b.

(ii) Find the equation of a line through the point $A(4, -3)$ having the same slope as line $x + 4y - 1 = 0$

10. (a) A doctor advice the patient to take 500 mg of calcium and 1,200 micrograms of vitamin A by drinking milk and orange juice. A unit of milk contains 38 mg of calcium and 56 micrograms of vitamin A while a unit of orange juice contains 5 mg of calcium and 60 micrograms of vitamin A. How many units of each should the patient take?

(b) Given the set numbers, $-5, 0, 12, -3.416520 \ldots, 2.85, 7.14, 12.64646464, \frac{11}{5}, -\frac{3}{5}, -2\frac{1}{3}$. state which of these numbers are;

(i) Positive rational numbers (ii) Negative rational number (iii) Irrational numbers

(c) (i) The base of a lid of bucket has an area of $\frac{25}{485}\pi m^2$. Find its diameter in decimal.

(ii) Represent $-4 \leq x < 4$ on a number line, where x is a real number.

(d) Solve the following simultaneous equations by using elimination and substitution methods

$$\begin{cases} \frac{4}{x} + \frac{1}{y} = 3 \\ \frac{1}{x} + \frac{2}{y} = -1 \end{cases}$$

**********************ALL THE BEST**********************

''Approximations are used in various real-life situations like estimating cost of items, distances, and ingredients for a recipe.''

@ BN SHARIFF OMAR